Intro to General Relativity
Homework #1

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Problem 1 (Schutz Ch.1 #5)

Problem Statement

(a) Use the spacetime diagram of an observer O to describe the following experiment performed by O. Two bursts of particles of speed $v = 0.5$ are emitted from $x = 0$ at $t = -2\,\mathrm{m}$, one traveling in the positive x direction and the other in the negative x direction. These encounter detectors located at $x = \pm 2\,\mathrm{m}$. After a delay of $0.5\,\mathrm{m}$ of time, the detectors send signals back to $x = 0$ at speed $v = 0.75$.

(b) The signals arrive back at $x = 0$ at the same event. From this the experimenter concludes that the particle detectors did indeed send out their signals simultaneously, since he knows they are equal distances from $x = 0$. Explain why this conclusion is valid.

(c) A second observer $\tilde{O}$ moves with speed $v = 0.75$ in the negative x direction relative to O. Draw the spacetime diagram of $\tilde{O}$ and in it depict the experiment performed by O. Does $\tilde{O}$ conclude that the particle detectors sent out their signals simultaneously? If not, which signal was sent first?

(d) Compute the interval Δs^2 between the events at which the detectors emitted their signals, using both the coordinates of O and those of $\tilde{O}$.

Solution

Units: $c = 1$.

(a) Spacetime diagram in O

Particles are emitted from $x = 0$ at

$$t = -2.$$

Detector positions:

$$x = \pm 2.$$

Particle speed:

$$v_p = 0.5.$$

Travel time to detector:

$$\Delta t_p = \frac{2}{0.5} = 4.$$

Arrival time at detectors:

$$t_{\text{hit}} = -2 + 4 = 2.$$

Each detector waits 0.5:

$$t_{\text{emit}} = 2 + 0.5 = 2.5.$$

Emission events:

$$E_+ : (t, x) = (2.5, +2), \qquad E_- : (t, x) = (2.5, -2).$$

Return signal speed:

$$v_s = 0.75.$$

Return travel time:

$$\Delta t_s = \frac{2}{0.75} = \frac{8}{3}.$$

Arrival at origin:

$$t_{\text{arr}} = 2.5 + \frac{8}{3} = \frac{31}{6}.$$

Both signals arrive at

$$\left(\frac{31}{6}, 0\right).$$

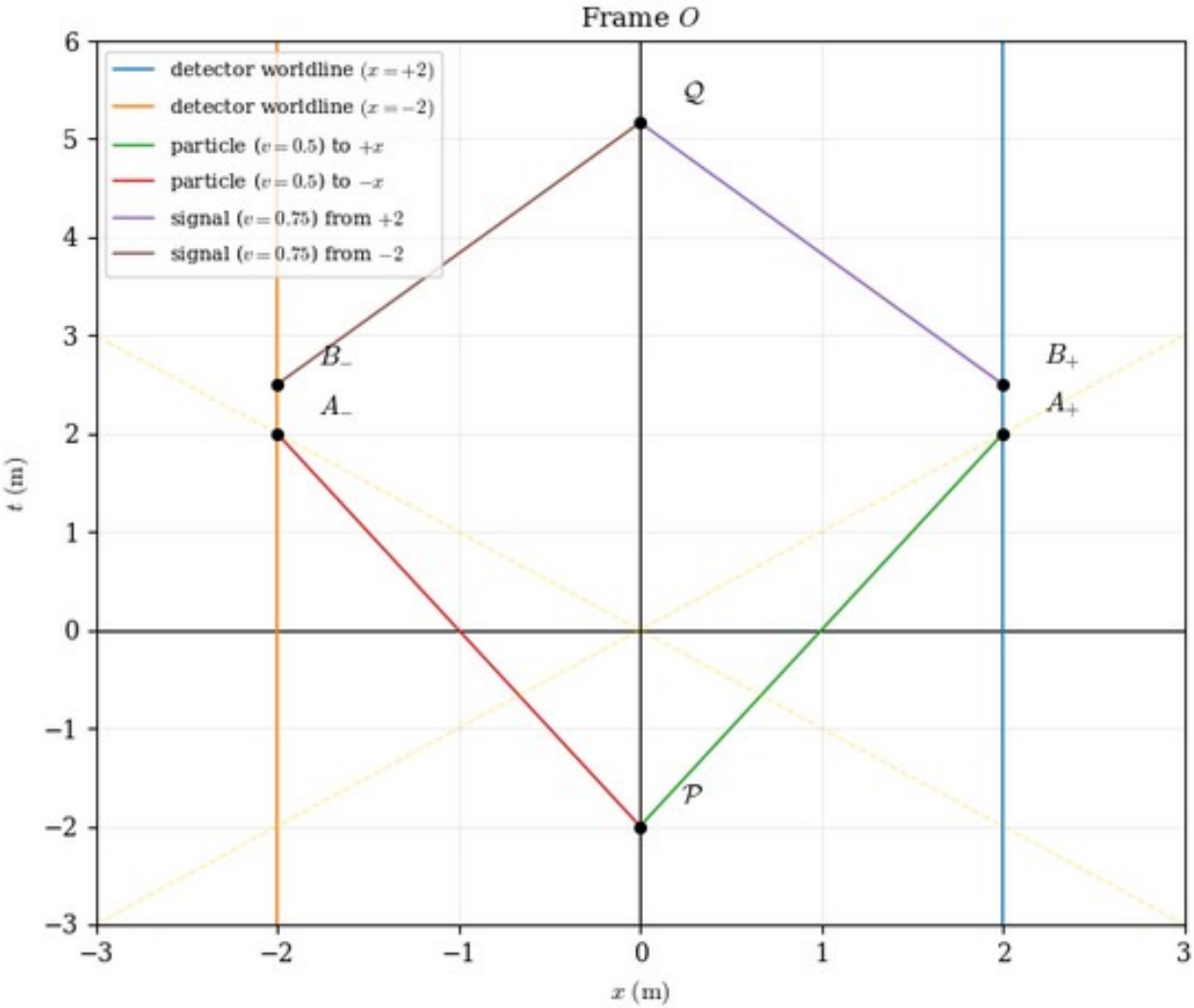

Figure 1: Spacetime diagram in O.

(b) The detectors are equidistant from $x = 0$ and the return signals propagate at the same speed. Since both signals arrive at the same event at $x = 0$, the emission times must be equal in O.

$$t_+(O) = t_-(O) = 2.5.$$

(c) Observer $\tilde{O}$ with $v = -0.75$ relative to O

Lorentz transformation:

$$t' = \gamma(t - vx), \qquad x' = \gamma(x - vt), \qquad \gamma = \frac{1}{\sqrt{1 - v^2}}.$$

With $v = -0.75$:

$$\gamma = \frac{4}{\sqrt{7}}.$$

Emission events:

$$t'_+ = \gamma(2.5 - v(2)), \qquad t'_- = \gamma(2.5 - v(-2)).$$

Substitute $v = -0.75$:

$$t'_+ = \gamma(4), \qquad t'_- = \gamma(1).$$

$$t'_+ \neq t'_-.$$

The emission from $x = -2$ occurs first in $\tilde{O}$.

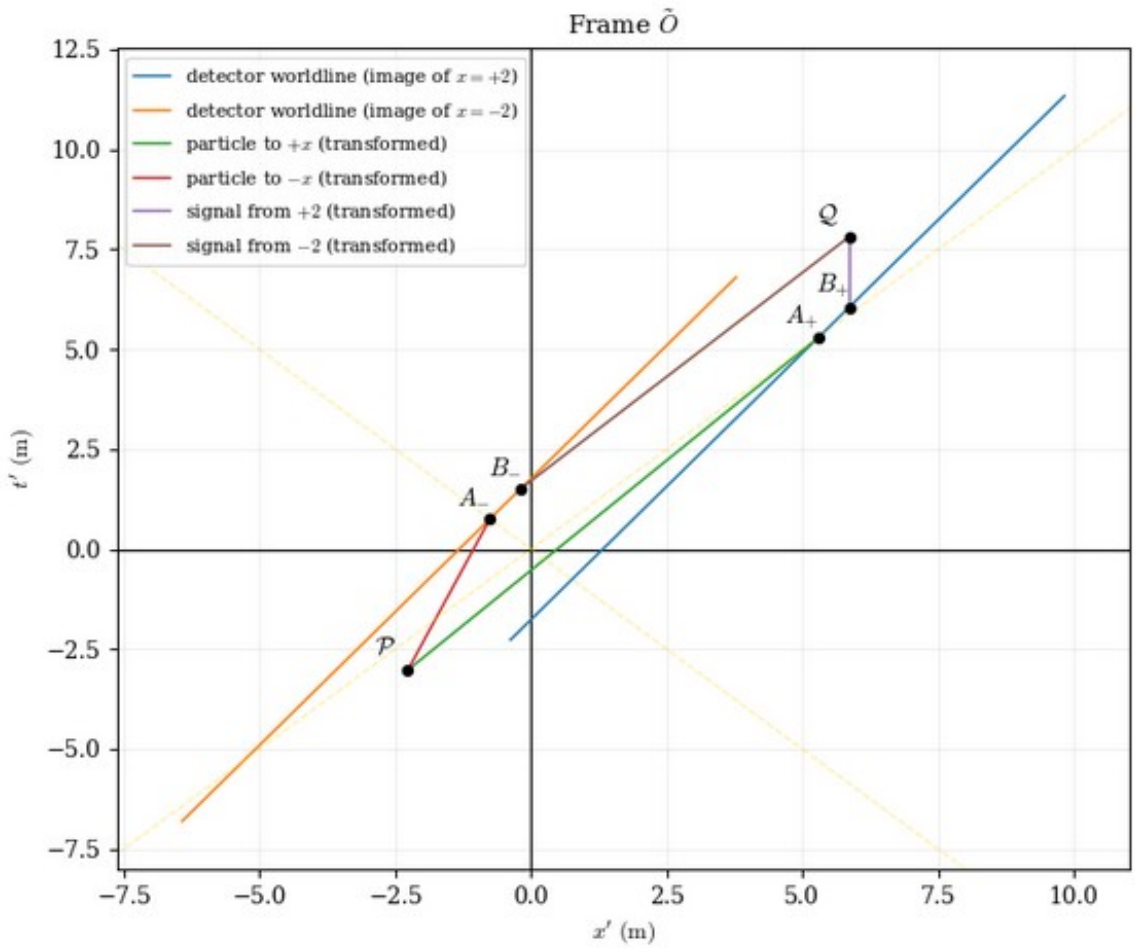

Figure 2: Spacetime diagram in $\tilde{O}$ coordinates.

(d) Interval between emission events

In O:

$$\Delta t = 0, \qquad \Delta x = 4.$$

$$\Delta s^2 = -(\Delta t)^2 + (\Delta x)^2 = 16.$$

In $\tilde{O}$:

$$\Delta t' = 3\gamma, \qquad \Delta x' = 4\gamma.$$

$$\Delta s^2 = -(3\gamma)^2 + (4\gamma)^2 = 7\gamma^2 = 16.$$

$$\boxed{\Delta s^2 = 16}$$

Problem 2 (Schutz Ch.1 #17)

Problem Statement

A lightweight pole 20 m long lies on the ground next to a barn 15 m long. An Olympic athlete picks up the pole, carries it far away, and runs with it toward the end of the barn at a speed $0.8c$. His friend remains at rest, standing by the door of the barn. Attempt all parts of this question, even if you can't answer some.

(a) How long does the friend measure the pole to be, as it approaches the barn?

(b) The barn door is initially open, and immediately after the runner and pole are entirely inside the barn, the friend shuts the door. How long after the door is shut, as measured by the friend, does the front of the pole hit the other end of the barn? Compute the interval between the events of shutting the door and hitting the wall. Is it spacelike, timelike, or null?

(c) In the reference frame of the runner, what is the length of the barn and the pole?

(d) Does the runner believe that the pole is entirely inside the barn when its front hits the end of the barn? Can you explain why?

(e) After the collision, the pole and runner come to rest relative to the barn. From the friend's point of view, the 20 m pole is now inside a 15 m barn, since the barn door was shut before the pole stopped. How is this possible? Alternatively, from the runner's point of view, the collision should have stopped the pole before the door closed, so the door could not be closed at all. Was or was not the door closed with the pole inside?

(f) Draw a spacetime diagram from the friend's point of view and use it to illustrate and justify all your conclusions.

Solution

Pole proper length:

$$L_{\text{pole}} = 20\,\text{m},$$

Barn proper length:

$$L_{\text{barn}} = 15\,\text{m}.$$

Runner speed $v = 0.8$, so

$$\gamma = \frac{1}{\sqrt{1-v^2}} = \frac{1}{\sqrt{1-0.64}} = \frac{5}{3}.$$

(a) In the barn frame, $\mathcal{O}$, the pole is length contracted:

$$L'_{\text{pole}} = \frac{L_{\text{pole}}}{\gamma} = \frac{20}{5/3} = 12\,\text{m}.$$

(b) In $\mathcal{O}$, let

$$\mathcal{B} = \text{rear of pole reaches door (door closes)},$$
$$\mathcal{C} = \text{front of pole reaches back wall}.$$

The rear reaches the door when the pole has advanced its contracted length:

$$t_{\mathcal{B}} = \frac{12}{0.8} = 15.$$

The front reaches the back wall at

$$t_{\mathcal{C}} = \frac{15}{0.8} = 18.75.$$

$$\Delta t = t_{\mathcal{C}} - t_{\mathcal{B}} = 18.75 - 15 = 3.75.$$

The spatial separation is

$$\Delta x = 15.$$

The invariant interval is

$$\Delta s^2 = -(\Delta t)^2 + (\Delta x)^2 = -(3.75)^2 + 15^2 = -14.0625 + 225 = 210.9375.$$

Since $\Delta s^2 > 0$, the separation is spacelike.

(c) In the runner's frame $\tilde{\mathcal{O}}$ the pole has its proper length:

$$L_{\text{pole}} = 20\,\text{m}.$$

The barn is length contracted:

$$L'_{\text{barn}} = \frac{L_{\text{barn}}}{\gamma} = \frac{15}{5/3} = 9\,\text{m}.$$

(d) No. In the runner's frame the barn is only 9 m long while the pole is 20 m long. The collision with the back wall occurs before the door closes. The events in part (b) are spacelike separated, so their time order depends on the frame.

(e) In the barn frame the door is closed when the entire contracted pole lies between $x = 0$ and $x = 15$.

In the runner's frame the collision occurs before the door closes. Because the events are spacelike separated, this difference in ordering does not violate causality. In $\mathcal{O}$ the door is closed with the pole inside, but not in $\tilde{\mathcal{O}}$.

(f)

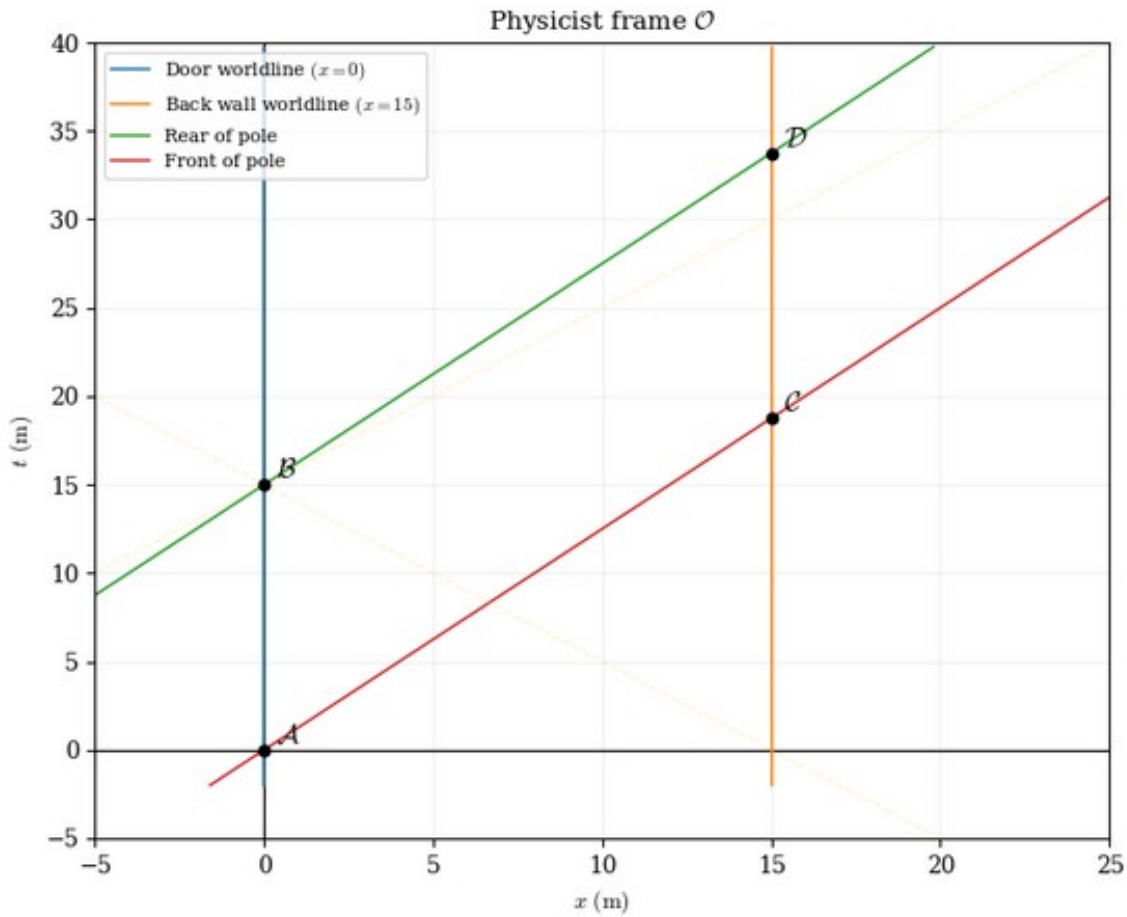

Figure 3: Spacetime diagram in the barn frame $\mathcal{O}$

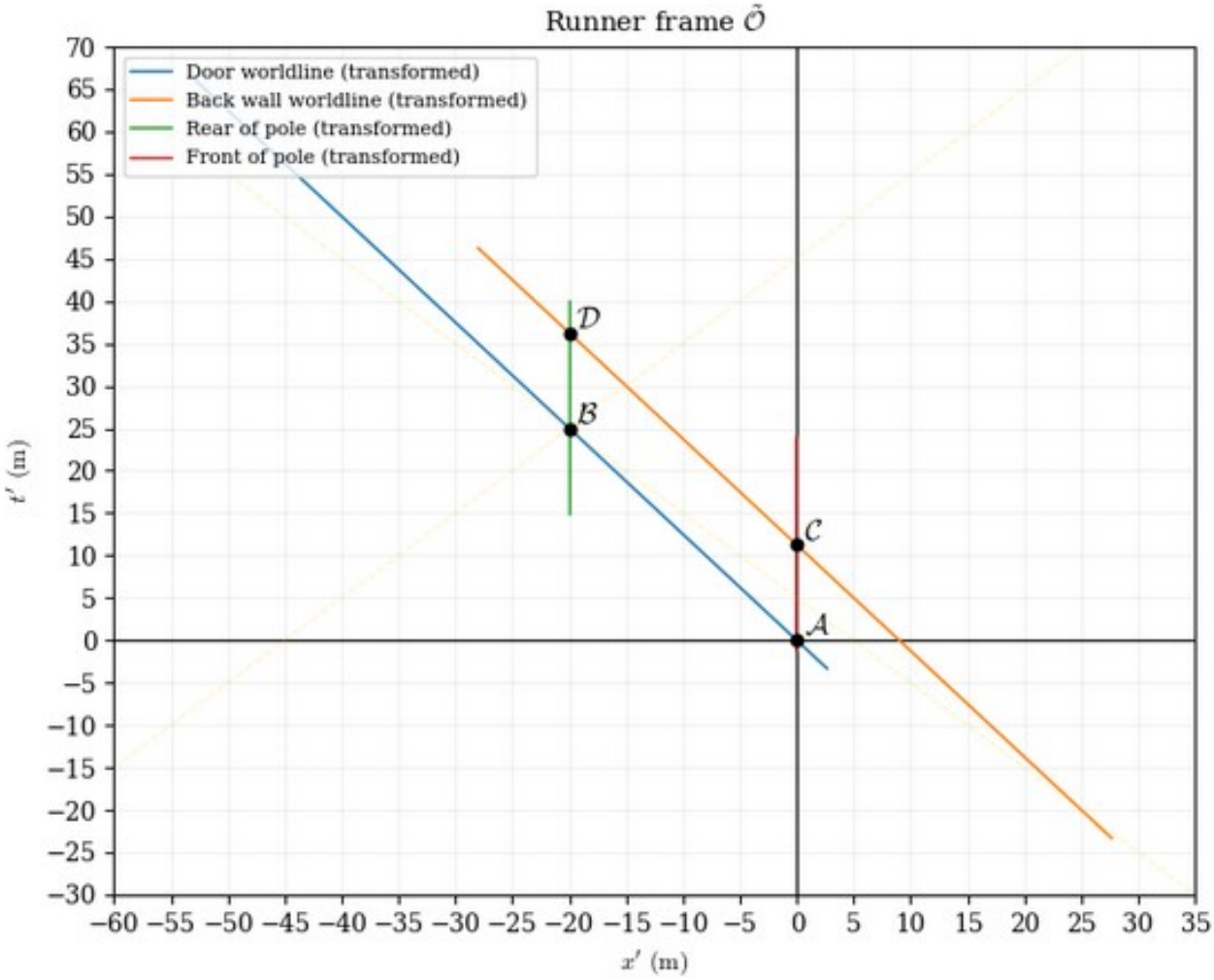

Figure 4: Spacetime diagram in the runner's frame $\tilde{\mathcal{O}}$

The diagrams show that events $\mathcal{B}$ and $\mathcal{C}$ are spacelike separated and therefore may reverse time order between frames.

Problem 3 (Schutz Ch.1 #19)

Problem Statement

The Einstein velocity-addition law, Eq. (1.13), has a simpler form if we introduce the velocity parameter u, defined by

$$v = \tanh u.$$

Notice that for $-\infty < u < \infty$, the velocity is confined to the acceptable limits $-1 < v < 1$. Show that if

$$v = \tanh u, \qquad w = \tanh U,$$

then Eq. (1.13) implies

$$w' = \tanh(u + U).$$

This means that velocity parameters add linearly.

(a) Using the velocity parameter introduced in Exer. 18, show that the Lorentz transformation equations, Eq. (1.12), can be written

$$\bar{t} = t \cosh u - x \sinh u, \qquad \bar{y} = y$$

$$\bar{x} = -t \sinh u + x \cosh u, \qquad \bar{z} = z.$$

(b) Use the identity $\cosh^2 u - \sinh^2 u = 1$ to demonstrate the invariance of the interval.

(c) Draw as many parallels as you can between the geometry of spacetime and ordinary two-dimensional Euclidean geometry, where the coordinate transformation analogous to the Lorentz transformation is

$$\bar{x} = x \cos\theta + y \sin\theta,$$

$$\bar{y} = -x \sin\theta + y \cos\theta.$$

What is the analog of the interval? Of the invariant hyperbola?

Solution

(a) Standard Lorentz transformation (Eq. 1.12):

$$\bar{t} = \gamma(t - vx), \qquad \bar{x} = \gamma(x - vt), \qquad \bar{y} = y, \qquad \bar{z} = z.$$

Let

$$v = \tanh u.$$

$$\gamma = \frac{1}{\sqrt{1 - v^2}} = \frac{1}{\sqrt{1 - \tanh^2 u}} = \frac{1}{\sqrt{\operatorname{sech}^2 u}} = \cosh u.$$

$$\gamma v = \cosh u \, \tanh u = \cosh u \, \frac{\sinh u}{\cosh u} = \sinh u.$$

Then

$$\bar{t} = \gamma t - \gamma v x = (\cosh u)\, t - (\sinh u)\, x,$$

$$\bar{x} = \gamma x - \gamma v t = (\cosh u)\, x - (\sinh u)\, t,$$

$$\bar{y} = y, \qquad \bar{z} = z.$$

$$\boxed{\bar{t} = t \cosh u - x \sinh u,}$$

$$\boxed{\bar{x} = -t \sinh u + x \cosh u.}$$

(b) Interval

$$\Delta s^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2.$$

$$\Delta\bar{t} = \Delta t \cosh u - \Delta x \sinh u, \qquad \Delta\bar{x} = -\Delta t \sinh u + \Delta x \cosh u, \qquad \Delta\bar{y} = \Delta y, \qquad \Delta\bar{z} = \Delta z.$$

$$-(\Delta\bar{t})^2 + (\Delta\bar{x})^2 = -(\Delta t \cosh u - \Delta x \sinh u)^2 + (-\Delta t \sinh u + \Delta x \cosh u)^2$$

$$= -\Big[(\Delta t)^2 \cosh^2 u - 2(\Delta t)(\Delta x) \cosh u \, \sinh u + (\Delta x)^2 \sinh^2 u\Big] + ...$$
$$\Big[(\Delta t)^2 \sinh^2 u - 2(\Delta t)(\Delta x) \sinh u \, \cosh u + (\Delta x)^2 \cosh^2 u\Big]$$

$$= -(\Delta t)^2(\cosh^2 u - \sinh^2 u) + (\Delta x)^2(\cosh^2 u - \sinh^2 u).$$

$$\cosh^2 u - \sinh^2 u = 1$$

$$-(\Delta\bar{t})^2 + (\Delta\bar{x})^2 = -(\Delta t)^2 + (\Delta x)^2.$$

$$\Delta\bar{s}^2 = -(\Delta\bar{t})^2 + (\Delta\bar{x})^2 + (\Delta\bar{y})^2 + (\Delta\bar{z})^2$$

$$= \Big[-(\Delta t)^2 + (\Delta x)^2\Big] + (\Delta y)^2 + (\Delta z)^2 = \Delta s^2.$$

$$\boxed{\Delta\bar{s}^2 = \Delta s^2.}$$

(c) Analog of the spacetime interval:

Euclidean rotation by angle θ:

$$\bar{x} = x\cos\theta + y\sin\theta, \qquad \bar{y} = -x\sin\theta + y\cos\theta.$$

$$(\Delta\bar{x})^2 + (\Delta\bar{y})^2 = (\Delta x)^2 + (\Delta y)^2.$$

Lorentz transformation in the t–x plane with parameter u:

$$\bar{t} = t\cosh u - x\sinh u, \qquad \bar{x} = -t\sinh u + x\cosh u.$$

$$-(\Delta\bar{t})^2 + (\Delta\bar{x})^2 = -(\Delta t)^2 + (\Delta x)^2.$$

Analog of invariant hyperbolae:

Euclidean circles:

$$x^2 + y^2 = \text{const.}$$

Invariant hyperbolae:

$$-t^2 + x^2 = \text{const.}$$

Problem 4 (Schutz Ch.2 #14)

Problem Statement

The following matrix gives a Lorentz transformation from $\mathcal{O}$ to $\bar{\mathcal{O}}$:

$$\left[\Lambda^{\bar{a}}{}_{b}\right] = \begin{pmatrix} 1.25 & 0 & 0 & 0.75 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0.75 & 0 & 0 & 1.25 \end{pmatrix}$$

(a) What is the velocity (speed and direction) of $\bar{\mathcal{O}}$ relative to $\mathcal{O}$?

(b) Find the inverse matrix to the given one.

(c) Find the components in $\mathcal{O}$ of a vector $\vec{A} \xrightarrow{\bar{\mathcal{O}}} (1, 2, 0, 0)$.

Solution

(a) For a pure boost in the z-direction,

$$\bar{t} = \gamma(t - vz), \qquad \bar{z} = \gamma(z - vt)$$

so

$$\Lambda^{\bar{0}}{}_{0} = \gamma, \qquad \Lambda^{\bar{0}}{}_{3} = -\gamma v$$

From the matrix,

$$\gamma = 1.25, \qquad -\gamma v = 0.75$$

$$\boxed{v = -\frac{0.75}{1.25} = -0.6}$$

(b) Inverse boost means v goes to $-v$ making off-diagonal entries change sign:

$$\left[\Lambda^{a}{}_{\bar{b}}\right] = \begin{pmatrix} 1.25 & 0 & 0 & -0.75 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -0.75 & 0 & 0 & 1.25 \end{pmatrix}$$

(c) Components of $\vec{A} \xrightarrow{\bar{\mathcal{O}}} (1, 2, 0, 0)$

$$A^a = \Lambda^a{}_{\bar{b}}\, A^{\bar{b}}, \qquad \vec{A} \xrightarrow{\bar{\mathcal{O}}} (A^{\bar{0}}, A^{\bar{1}}, A^{\bar{2}}, A^{\bar{3}}) = (1, 2, 0, 0).$$

Then

$$A^0 = 1.25(1) - 0.75(0) = 1.25, \qquad A^1 = 2, \qquad A^2 = 0, \qquad A^3 = -0.75(1) + 1.25(0) = -0.75.$$

$$\boxed{\vec{A} \xrightarrow{\mathcal{O}} (1.25,\ 2,\ 0,\ -0.75).}$$

Problem 5 (Schutz Ch.2 #25)

Problem Statement

Let frame $\bar{\mathcal{O}}$ move with speed v in the x-direction relative to $\mathcal{O}$. Let a photon have frequency ν in $\mathcal{O}$ and move at an angle θ with respect to $\mathcal{O}$'s x axis.

(a) Show that its frequency in $\bar{\mathcal{O}}$ is
$$\frac{\bar{\nu}}{\nu} = \frac{1 - v\cos\theta}{\sqrt{1-v^2}}.$$

(b) Even when $\theta = \pi/2$ there is a frequency shift (transverse Doppler shift). What angle θ gives *no* Doppler shift between $\mathcal{O}$ and $\bar{\mathcal{O}}$?

Solution

(a) Measured frequency to measured photon energy relates via

$$E = h\nu, \qquad \bar{E} = h\bar{\nu},$$

so the problem reduces to finding how $p^0 = E$ changes under a boost from $\mathcal{O}$ to $\bar{\mathcal{O}}$.

In $\mathcal{O}$ the photon moves at angle θ relative to the x-axis, so its spatial momentum points along

$$\hat{\mathbf{n}} = (\cos\theta, \sin\theta, 0),$$

and for a photon $|\mathbf{p}| = E$. Thus

$$\vec{p} \xrightarrow{\mathcal{O}} (p^0, p^1, p^2, p^3) = (E,\ E\cos\theta,\ E\sin\theta,\ 0).$$

A boost in the x-direction mixes only the $(0,1)$ components (time and x), leaving p^2, p^3 unchanged. With
$$\gamma = \frac{1}{\sqrt{1-v^2}},$$
the transformation of the time-component is

$$\bar{p}^0 = \gamma\,(p^0 - vp^1).$$

Substituting the components above,

$$\bar{E} = \bar{p}^0 = \gamma\big(E - v(E\cos\theta)\big) = \gamma E(1 - v\cos\theta).$$

Using $E = h\nu$ and $\bar{E} = h\bar{\nu}$,

$$h\bar{\nu} = \gamma(h\nu)(1 - v\cos\theta)$$
$$\frac{\bar{\nu}}{\nu} = \gamma(1 - v\cos\theta).$$

Insert γ:

$$\boxed{\frac{\bar{\nu}}{\nu} = \frac{1 - v\cos\theta}{\sqrt{1 - v^2}}.}$$

(b) Assign the same frequency because there is no Doppler shift:

$$\bar{\nu} = \nu \quad \Longleftrightarrow \quad \frac{\bar{\nu}}{\nu} = 1.$$

Using the result of (a),

$$\gamma(1 - v\cos\theta) = 1.$$

Since $\gamma^{-1} = \sqrt{1 - v^2}$,

$$1 - v\cos\theta = \sqrt{1 - v^2}.$$

Then,

$$v\cos\theta = 1 - \sqrt{1 - v^2}$$
$$\boxed{\cos\theta = \frac{1 - \sqrt{1 - v^2}}{v}}$$

So the direction of theta:

$$\boxed{\theta = \arccos\left(\frac{1 - \sqrt{1 - v^2}}{v}\right)}$$

For $\theta = \pi/2$,

$$\frac{\bar{\nu}}{\nu} = \gamma(1 - v\cos\theta) = \gamma,$$

so a frequency shift remains.